

Bluestock Mutual Fund Analytics Capstone

End-to-End Mutual Fund Analytics and Business Intelligence Solution

Submitted By

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Internship Organization

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Project Duration

May 2026 – June 2026

Technologies Used

Python | Pandas | NumPy | SQLite | SQLAlchemy | Jupyter Notebook | Power BI | Git | GitHub

Submission Date

June 2026

Executive Summary:

The Mutual Fund Analytics Capstone Project was undertaken to develop a comprehensive data analytics solution for analyzing mutual fund performance, investor behavior, industry trends, and risk metrics. The project demonstrates the complete lifecycle of a data analytics workflow, beginning with raw data ingestion and culminating in an interactive business intelligence dashboard.

The mutual fund industry generates significant volumes of data across fund performance, investor transactions, asset management, portfolio holdings, and market benchmarks. Extracting meaningful insights from such data requires a structured approach involving data engineering, statistical analysis, financial modeling, and visualization. This project was designed to address that challenge by building an end-to-end analytics platform capable of transforming raw financial data into actionable business insights.

The implementation consisted of multiple stages. Initially, ten mutual fund datasets were ingested and validated using automated ETL pipelines developed in Python. Data quality checks were performed to identify missing values, duplicate records, inconsistent formats, and invalid entries. The cleaned datasets were subsequently stored within a SQLite analytical database designed using a star schema architecture.

Exploratory Data Analysis (EDA) was performed to understand industry trends, investor demographics, category inflows, and fund behavior. Multiple visualizations were created to identify growth patterns, category preferences, regional investment trends, and SIP participation across the analysis period.

Performance analytics were conducted using key financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and benchmark comparison measures. These metrics enabled the evaluation of mutual fund performance from both return and risk perspectives.

Advanced analytics extended the analysis through Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio analysis, investor cohort analysis, SIP continuity analysis, and portfolio concentration assessment using the Herfindahl-Hirschman Index (HHI).

Finally, an interactive Power BI dashboard was developed to provide decision-makers with a consolidated view of industry trends, fund performance, investor behavior, and SIP growth. The dashboard supports dynamic filtering, drill-through navigation, and visual exploration of key performance indicators.

The project successfully demonstrates the application of data engineering, business analytics, and financial analysis techniques within the mutual fund domain and provides actionable insights for investors, fund managers, and financial analysts.

Problem Statement and Objectives:

Problem Statement:

The Indian mutual fund industry has witnessed substantial growth over the past decade, resulting in increased participation from retail and institutional investors. As the industry expands, the volume and complexity of data generated by mutual fund schemes continue to increase. This includes information related to fund performance, investor transactions, portfolio allocations, assets under management, benchmark indices, and investor demographics.

While large volumes of data are available, deriving meaningful insights from this information remains a challenge. Investors require tools to evaluate risk-adjusted returns, compare funds against benchmarks, understand market trends, and identify suitable investment opportunities. Similarly, fund managers and financial institutions require analytical frameworks to monitor investor behavior, assess fund performance, and identify emerging trends within the market.

This project aims to address these challenges by developing a complete analytics platform capable of ingesting, processing, analyzing, and visualizing mutual fund data.

Project Objectives:

The primary objectives of this project are:

1. Develop an automated ETL pipeline for mutual fund datasets.
2. Design and implement an analytical database using SQLite.
3. Perform comprehensive data cleaning and validation.
4. Conduct exploratory data analysis to identify industry trends.
5. Evaluate mutual fund performance using financial performance metrics.
6. Analyze investor demographics and transaction behavior.
7. Implement advanced risk analytics and recommendation logic.
8. Develop an interactive Power BI dashboard for business users.
9. Generate actionable insights and recommendations for decision-making.
10. Demonstrate end-to-end analytics capabilities within the fintech domain.

Scope of the Project:

The scope of this project includes mutual fund analytics covering the period 2022–2026 using ten structured datasets. The analysis focuses on performance evaluation, investor behavior, SIP trends, risk assessment, benchmark comparison, and dashboard visualization. Real-time market execution and live portfolio management are outside the scope of this project.

Data Sources and Dataset Overview:

The project utilizes ten datasets representing different aspects of the mutual fund ecosystem. These datasets collectively provide information regarding fund performance, investor activity, assets under management, portfolio composition, and benchmark market movements.

Dataset Summary:

Dataset	Description
Fund Master	Master reference table containing scheme information and classifications
NAV History	Historical Net Asset Value records for mutual fund schemes
Investor Transactions	Investor purchase, SIP, redemption, and transaction records
Scheme Performance	Performance metrics including returns, Sharpe ratio, alpha, beta, and risk measures
AUM by Fund House	Assets under Management data for major fund houses
Monthly SIP Inflows	Industry SIP contributions and participation statistics
Category Inflows	Category-wise investment inflows and outflows
Industry Folio Count	Growth of investor folios across categories
Portfolio Holdings	Scheme-level portfolio allocation and sector exposure
Benchmark Indices	Historical market benchmark index values including NIFTY indices

Dataset Characteristics:

The datasets span the period from 2022 to 2026 and contain structured financial and transactional information. Together, they enable analysis across multiple dimensions including:

- Fund performance

- Risk assessment
- Investor demographics
- Market participation
- Portfolio composition
- Industry growth
- Benchmark comparison

Data Quality Considerations:

Prior to analysis, all datasets underwent a comprehensive validation process. The validation process included:

- Missing value detection
- Duplicate record identification
- Date standardization
- Numerical data validation
- AMFI code verification
- Referential integrity checks

These quality assurance measures ensured that subsequent analytical results were reliable and consistent.

Database Design:

A star schema architecture was implemented to support efficient analytical querying and dashboard integration. The design separates transactional data into fact tables while maintaining descriptive information within dimension tables.

Fact Tables

- fact_nav
- fact_transactions
- fact_performance
- fact_aum

Dimension Tables

- dim_fund
- dim_date

Observations

- The schema reduces data redundancy.
- Common analytical queries can be executed efficiently.
- The structure supports dashboard filtering and aggregation.

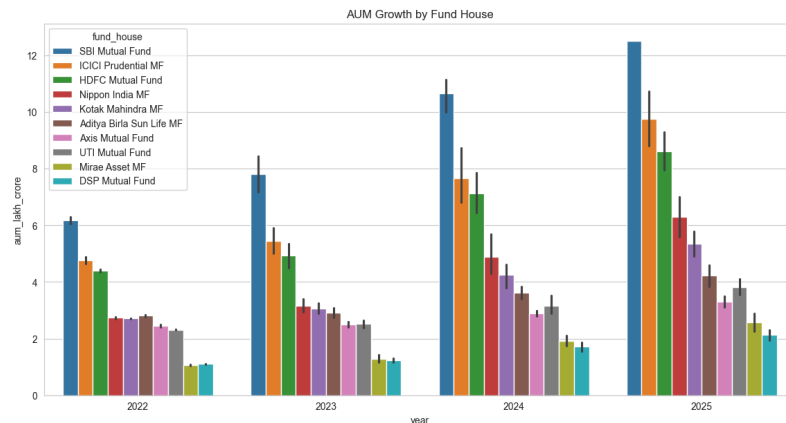
Business Interpretation

A star schema is widely used in data warehousing because it enables faster reporting and simplifies business intelligence workflows. The implemented design provides a scalable foundation for mutual fund analytics.

Exploratory Data Analysis: Industry Trends

Industry Growth Analysis:

The objective of this analysis was to understand the overall growth of the mutual fund industry between 2022 and 2025 using Assets Under Management (AUM) and SIP inflow trends.

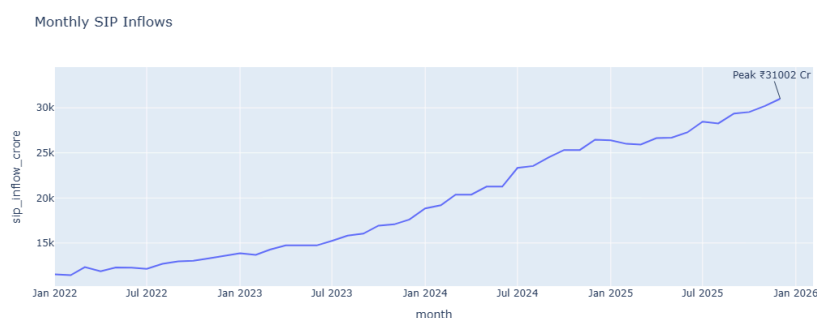


Observations

- Industry AUM increased consistently throughout the analysis period.
- Temporary slowdowns were observed during market correction phases.
- SBI Mutual Fund maintained leadership in total assets managed.

Business Interpretation

The sustained increase in AUM indicates growing investor confidence in mutual funds as a preferred investment vehicle. The trend also highlights the increasing role of retail investors in capital market participation.



Observations

- SIP inflows exhibited steady growth over time.
- The highest inflow was recorded in December 2025.
- SIP participation remained resilient during periods of market volatility.

Business Interpretation

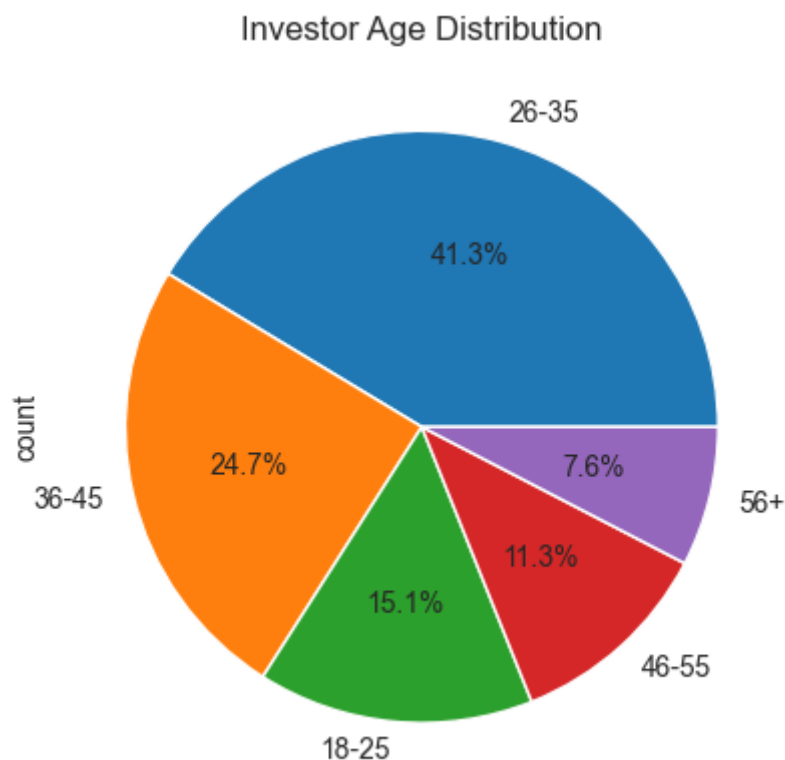
Investors increasingly adopted disciplined investing through SIPs, reinforcing the long-term investment culture within the mutual fund industry.

Exploratory Data Analysis: Investor Analytics

Investor Behaviour Analysis:

Investor demographic analysis was performed to understand participation patterns across age groups, genders, and geographic regions.

Age Group Distribution:



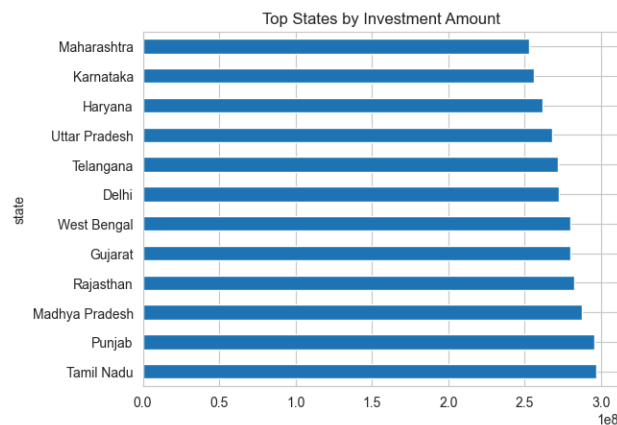
Observations

- Investors aged 26–45 represented the largest investor segment.
- Younger investors demonstrated strong participation through SIP investments.

Business Interpretation

The data suggests that mutual funds are increasingly becoming a preferred investment option among working professionals seeking long-term wealth creation.

State-wise Investment:



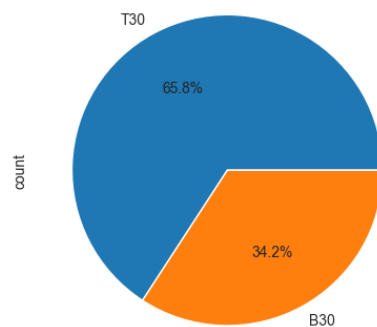
Observations

- Maharashtra, Karnataka, Telangana, and Delhi contributed a significant share of investment volume.
- Metropolitan regions dominated overall participation.

Business Interpretation

Investment activity remains concentrated in economically developed regions with higher financial literacy and digital adoption.

T30 vs B30 Analysis:



Observations

- T30 cities contributed the majority of investments.
- B30 participation continues to grow steadily.

Business Interpretation

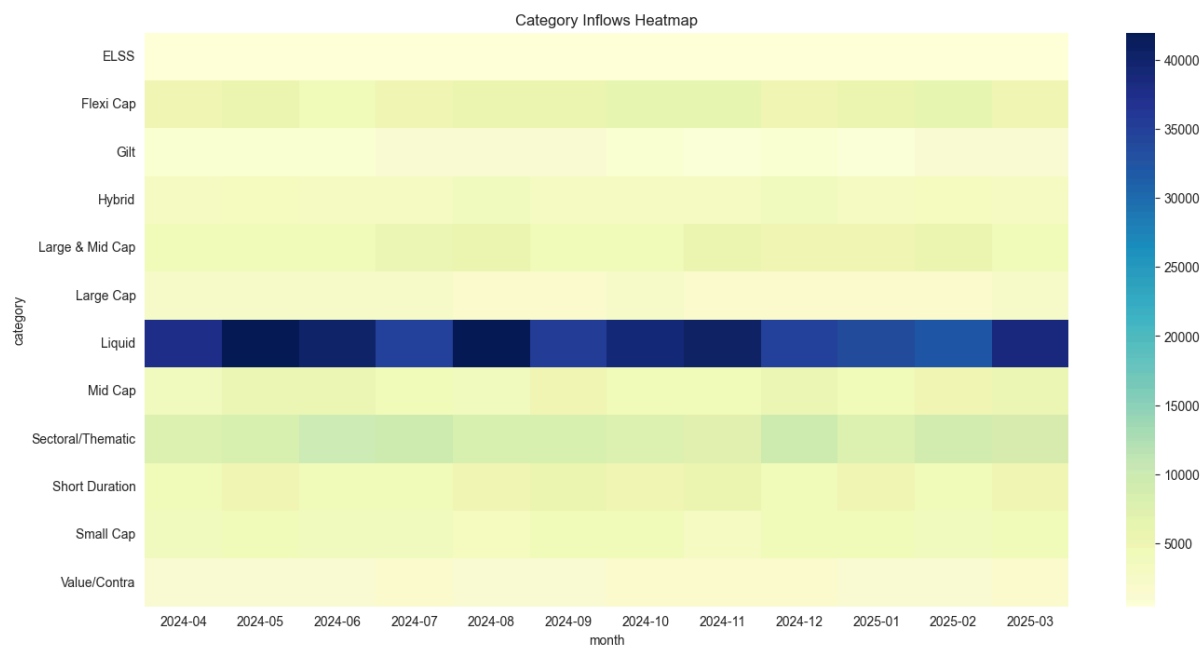
Mutual fund penetration is expanding beyond major metropolitan regions, indicating broader financial inclusion.

Exploratory Data Analysis: Category Trends

Category Inflow Analysis:

Category-level analysis was conducted to identify investor preferences across different mutual fund categories.

Category Inflow Heatmap:



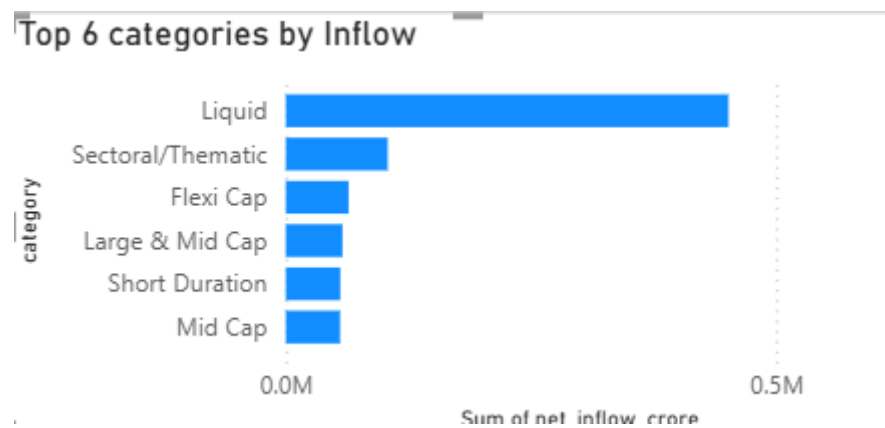
Observations

- The liquid category attracted significant inflows.
- Investor preferences remained mostly the same across cycles.

Business Interpretation

Investors demonstrated a willingness to pursue higher-growth categories despite increased risk levels.

Top Categories by Net Inflow:



Observations

- Equity-oriented categories consistently ranked among the highest inflow categories.
- Growth-oriented funds attracted strong investor interest.

Business Interpretation

The findings indicate a positive risk appetite among investors during the study period

Performance Analytics:

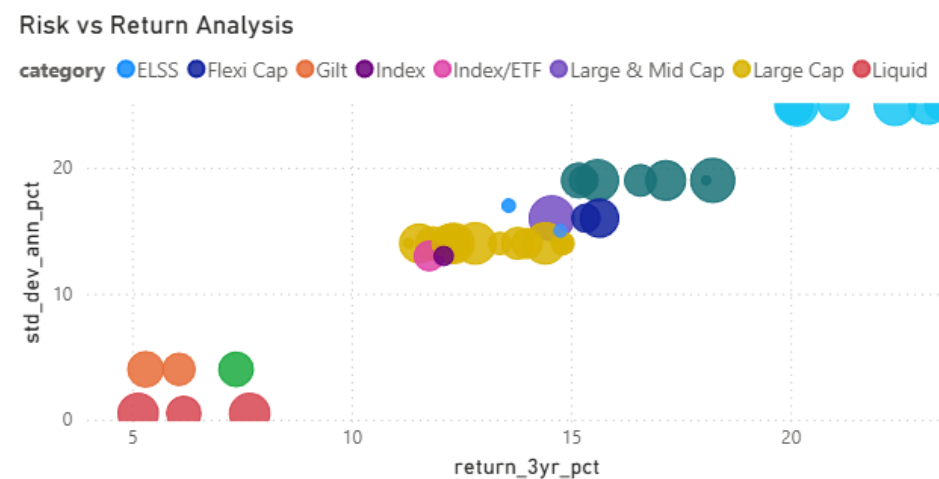
Mutual Fund Performance Evaluation

Performance evaluation was performed using risk-adjusted return metrics rather than relying solely on absolute returns.

Metrics Computed

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown

Risk vs Return Analysis



Observations

- Funds with higher returns generally exhibited higher volatility.
- Several funds achieved attractive risk-adjusted returns despite moderate risk levels.

Business Interpretation

Risk-adjusted performance provides a more meaningful measure of fund quality than absolute returns alone.

Fund Scorecard:

	amfi_code	scheme_name	category	plan	fund_score
0	119551	SBI Bluechip Fund - Regular Plan - Growth	Large Cap	Regular	50.567108
1	119552	SBI Bluechip Fund - Direct Plan - Growth	Large Cap	Direct	36.294896
2	119598	SBI Small Cap Fund - Regular Plan - Growth	Small Cap	Regular	17.580340
3	119599	SBI Small Cap Fund - Direct Plan - Growth	Small Cap	Direct	56.805293
4	119120	SBI Magnum Gilt Fund - Regular Plan - Growth	Gilt	Regular	82.419660

Observations

- A small group of funds consistently ranked highly across multiple performance indicators.
- Strong Sharpe ratios were associated with superior composite scores.

Business Interpretation

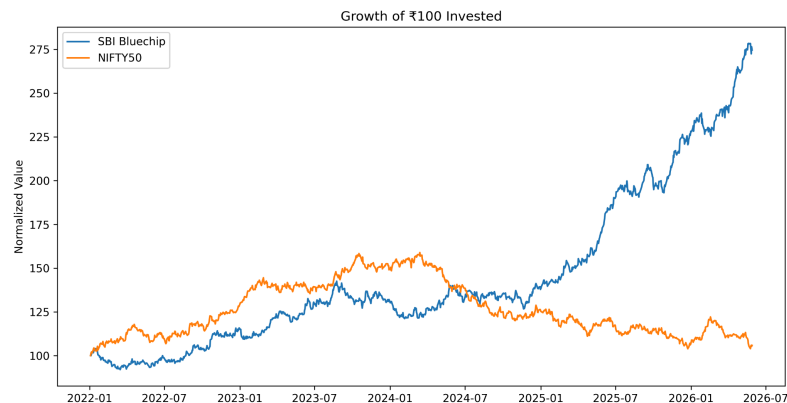
Investors should consider multiple performance dimensions when selecting mutual funds.

Benchmark Comparison

Fund Performance vs Benchmark Indices

Benchmark comparison was conducted using NIFTY 50 and NIFTY 100 indices.

NAV vs NIFTY 50:



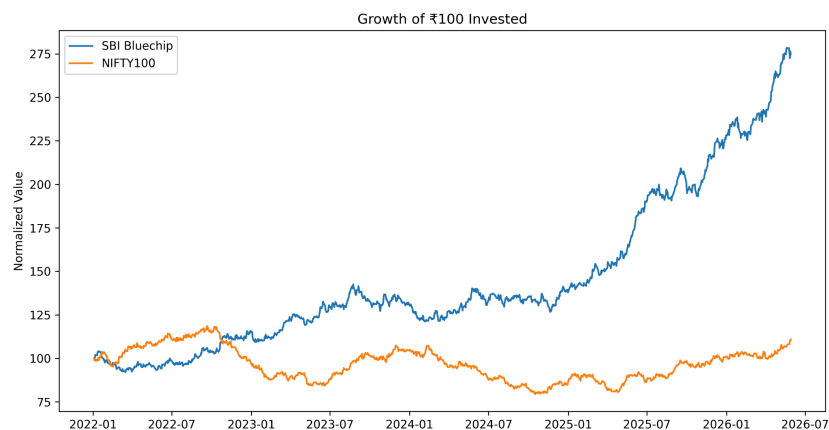
Observations

- Several funds outperformed benchmark indices during the analysis period.
- Active management generated measurable alpha in selected schemes.

Business Interpretation

Benchmark comparison helps investors assess whether fund managers are creating value beyond market returns.

NAV vs NIFTY100:



Observations

- Performance varied significantly across categories.
- Some funds delivered excess returns despite market volatility.

Business Interpretation

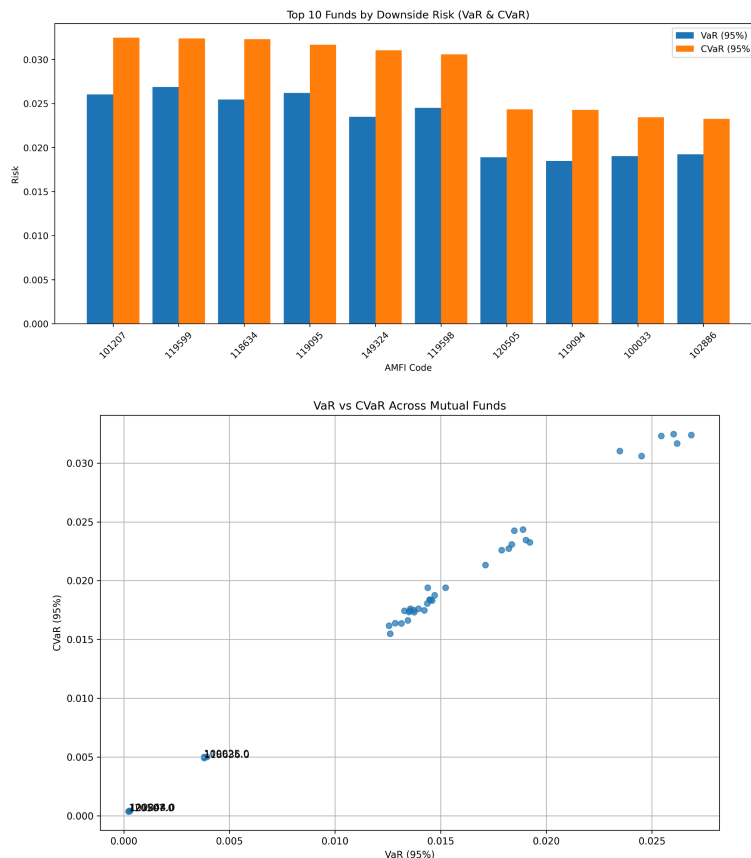
Outperformance relative to benchmarks indicates effective portfolio management and investment strategy execution.

Advanced Analytics: Risk Analysis

Risk Assessment and Downside Protection

Advanced risk metrics were computed to evaluate downside exposure and portfolio resilience

VaR and CVaR Comparison



Observation:

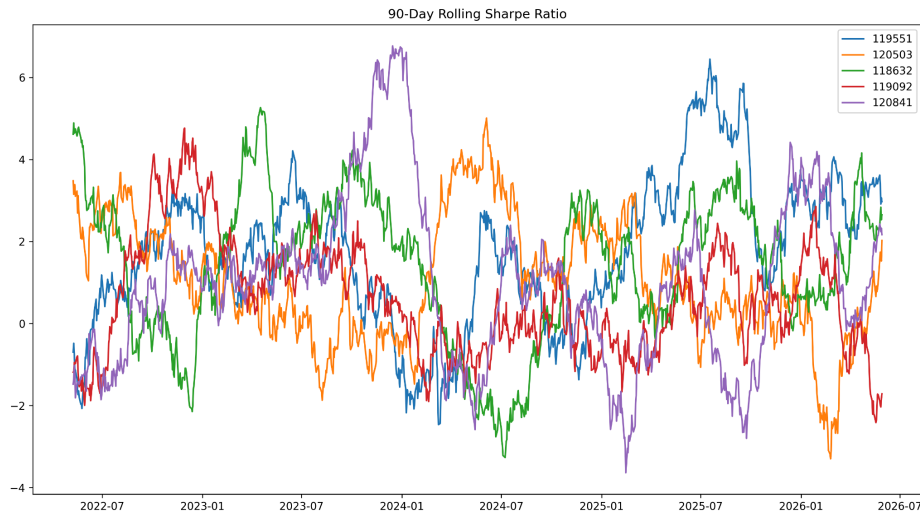
- Small-cap and aggressive equity funds exhibited the highest downside risk.
- CVaR values were consistently larger than VaR values, indicating that losses beyond the VaR threshold can be substantially more severe.
- Funds with high VaR also tended to exhibit high CVaR, suggesting greater exposure to tail-risk events.

Business Interpretation:

Value at Risk (VaR) estimates the maximum expected loss at a specified confidence level, while Conditional Value at Risk (CVaR) measures the average loss when that threshold is

exceeded. Together, these metrics provide a comprehensive assessment of downside risk and help investors select funds aligned with their risk tolerance.

Rolling 90-Day Sharpe Ratio:



Observations

- Risk-adjusted performance varied across market cycles.
- Market corrections led to temporary declines in Sharpe ratios.

Business Interpretation

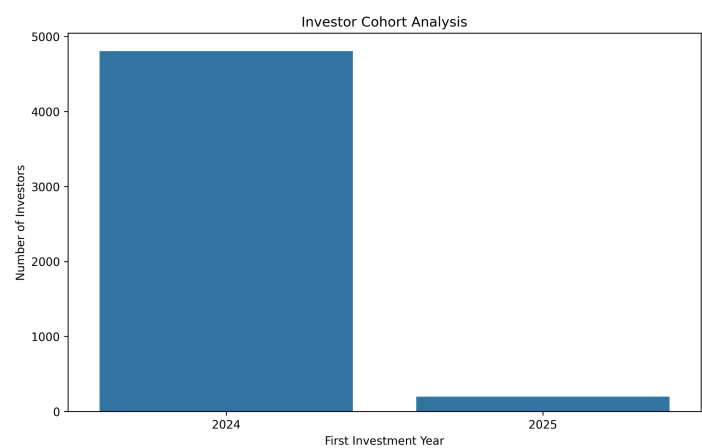
Rolling Sharpe analysis provides insight into changing performance dynamics over time.

Advanced Analytics: Investor Behaviour

Investor Cohort and SIP Analysis

Investor transaction data was analyzed to identify cohort behavior and SIP continuity patterns.

Investor Cohort Analysis:



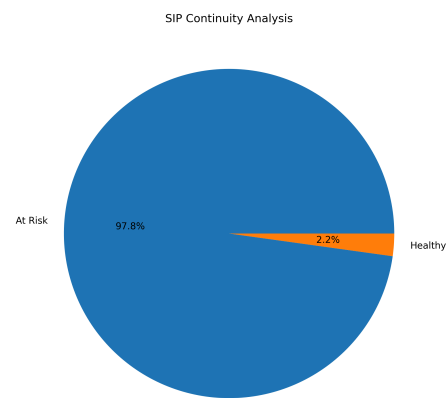
Observations

- Recent cohorts demonstrated lower participation levels.

Business Interpretation

Growing investor participation reflects increasing confidence in mutual funds as long-term investment vehicles.

SIP Continuity Analysis:



Observations

- Most investors maintained regular SIP schedules.
- We considered gap_days > 6 as a risk factor.
- A subset of investors exhibited irregular contribution patterns.

Business Interpretation

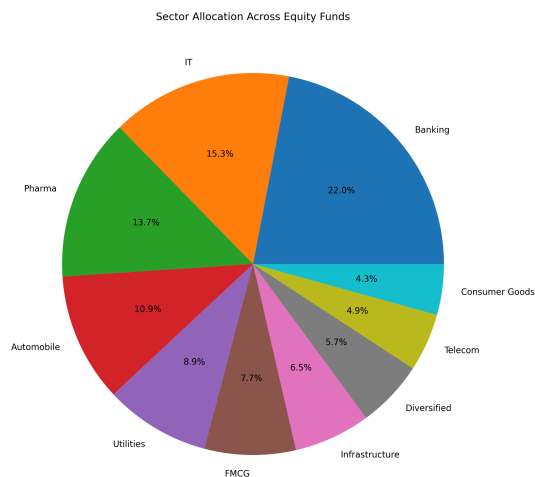
Identifying at-risk investors enables targeted engagement and retention strategies.

Portfolio Concentration Analysis

Portfolio Diversification Assessment

Sector allocation and portfolio concentration were analyzed using the Herfindahl-Hirschman Index (HHI).

Sector Allocation



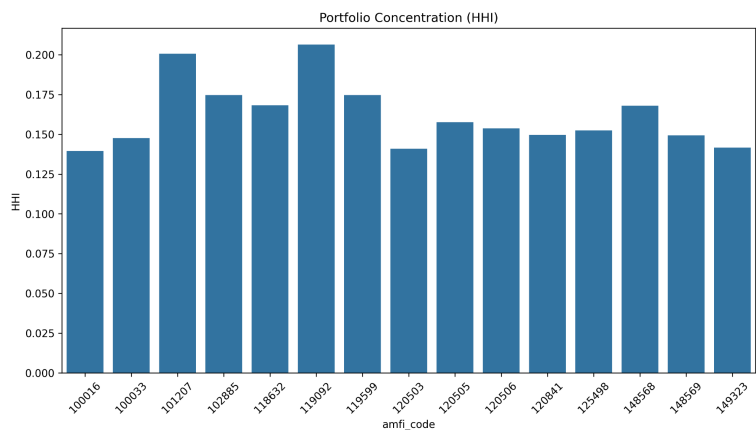
Observations

- Banking and Financial Services represented major allocations in several funds.
- Sector exposure varied across categories.

Business Interpretation

Sector concentration can significantly influence portfolio risk and return characteristics.

HHI Comparison:



Observations

- Some funds exhibited highly concentrated portfolios.
- Diversification levels varied considerably.

Business Interpretation

Investors should monitor concentration risk alongside traditional performance measures.

Dashboard Overview

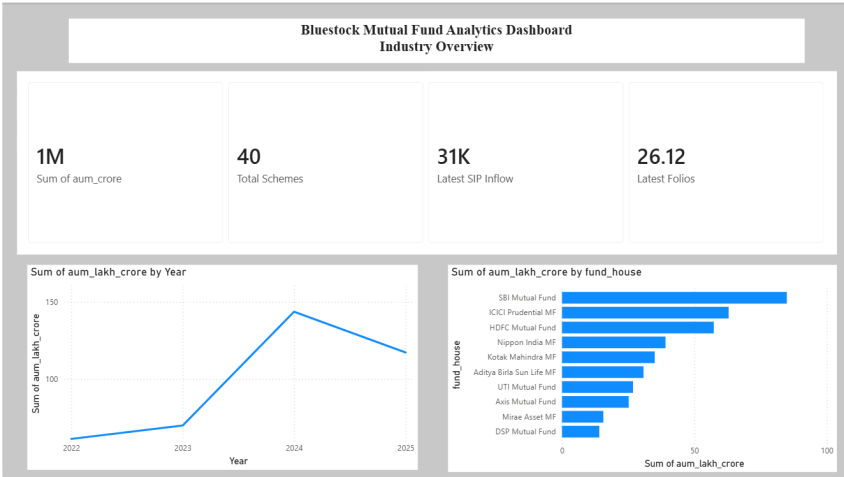
Interactive Business Intelligence Dashboard

A Power BI dashboard was developed to provide an integrated view of mutual fund performance, investor behavior, and industry trends.

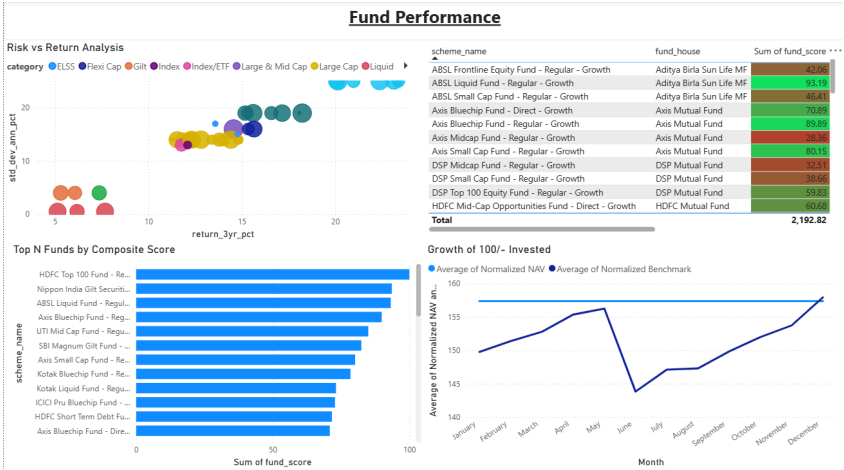
Dashboard Features

- Dynamic filtering
- Drill-through navigation
- KPI monitoring
- Benchmark comparison
- Investor analytics
- Category trend analysis

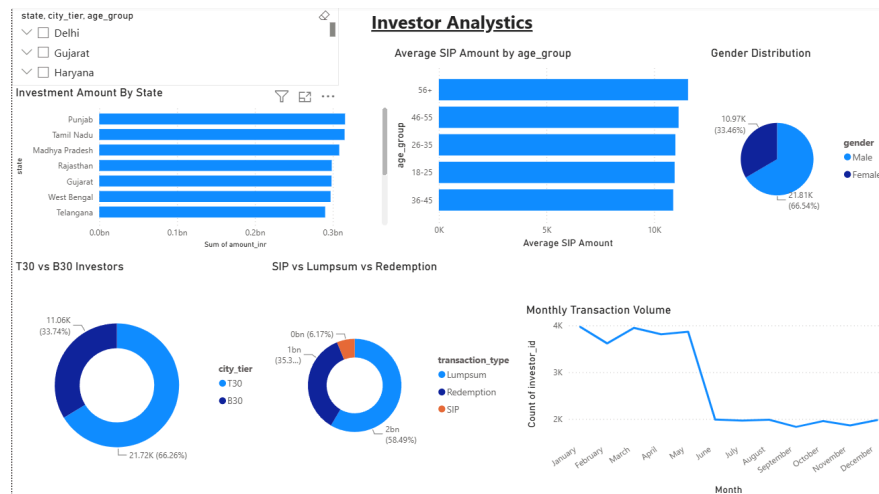
Industry Overview Dashboard:



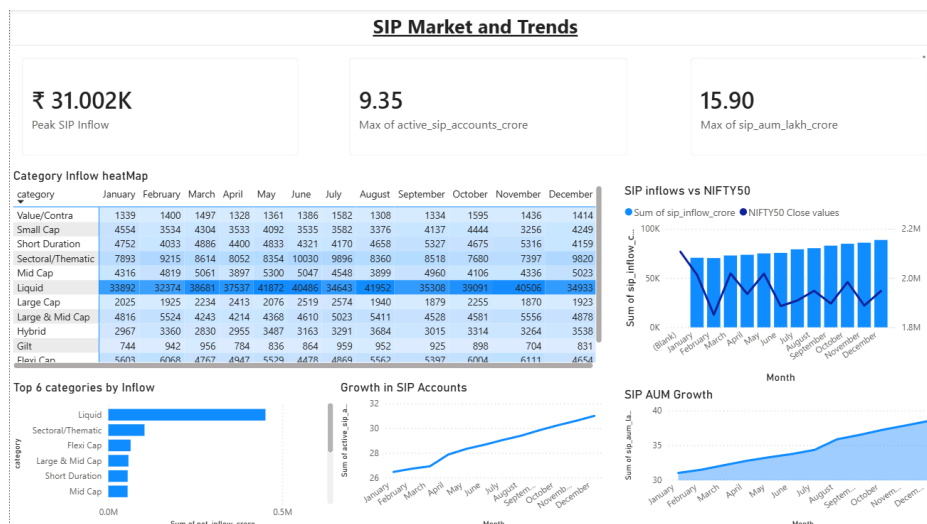
Fund Performance Dashboard



Investor Analytics Dashboard



SIP & Market Trends Dashboard



Business Value

The dashboard enables decision-makers to quickly explore data, identify trends, and monitor key performance indicators without requiring technical expertise.

Key Findings

The analysis of mutual fund industry datasets revealed several important insights regarding fund performance, investor behavior, industry growth, and risk characteristics.

1. Strong Industry Growth

The mutual fund industry demonstrated consistent growth in Assets Under Management (AUM) throughout the study period. Increasing investor participation and market expansion contributed significantly to overall industry growth.

2. SIP Investments Continue to Rise

Monthly SIP inflows exhibited a strong upward trend, reaching record highs by the end of the analysis period. This reflects increasing investor awareness and adoption of disciplined long-term investing strategies.

3. Small-Cap Funds Delivered Higher Returns but Higher Risk

Small-cap and mid-cap funds generally generated superior returns compared to large-cap funds. However, these funds also exhibited significantly higher volatility, VaR, CVaR, and drawdown risk.

4. Risk-Adjusted Metrics Provided Better Evaluation

Metrics such as Sharpe Ratio, Sortino Ratio, and Alpha provided a more meaningful evaluation of fund quality compared to absolute returns alone. Several funds with moderate returns achieved strong risk-adjusted performance.

5. Investor Participation Expanded Beyond Metropolitan Regions

While T30 cities contributed the majority of investments, B30 participation increased steadily during the study period, indicating broader mutual fund adoption across India.

6. Portfolio Concentration Varies Across Funds

HHI analysis revealed substantial variation in portfolio diversification. Certain funds exhibited concentrated sector exposure, increasing portfolio-specific risk.

7. Investor Cohorts Show Increasing Participation

Recent investor cohorts demonstrated higher average investment amounts and greater participation levels compared to earlier cohorts, reflecting growing confidence in mutual funds as an investment vehicle.

Recommendations

Based on the analytical findings, the following recommendations are proposed:

For Investors

- Focus on risk-adjusted returns rather than solely evaluating absolute performance.
- Consider diversification across multiple fund categories to reduce concentration risk.
- Maintain long-term SIP investments to benefit from disciplined wealth creation.
- Evaluate downside risk metrics such as VaR and Maximum Drawdown before investing.

For Fund Houses

- Strengthen investor education programs promoting SIP continuity.
- Expand outreach initiatives within B30 regions to increase participation.
- Monitor sector concentration and diversification levels within portfolios.
- Develop investor retention programs targeting at-risk SIP investors identified through continuity analysis.

For Business Analytics Teams

- Continuously monitor benchmark-relative performance.
- Incorporate predictive analytics for investor churn detection.
- Utilize cohort analysis for customer segmentation and personalized recommendations

Limitations

While the project successfully demonstrates end-to-end mutual fund analytics, several limitations should be acknowledged.

1. Dataset Scope

The analysis was performed using a representative mutual fund dataset and may not fully capture the entire mutual fund industry.

2. Historical Analysis

The project primarily focuses on historical performance. Future market behavior may differ significantly from historical trends.

3. Benchmark Assumptions

Alpha and beta calculations were performed using benchmark indices and assumed market relationships, which may vary under different market conditions.

4. Static Portfolio Holdings

Portfolio concentration analysis was based on available portfolio snapshots and may not reflect dynamic portfolio adjustments made by fund managers.

5. Simplified Recommendation Engine

The implemented recommendation system utilizes risk grades and Sharpe ratios. A production-grade recommendation engine would incorporate additional investor preferences and behavioral characteristics.

Conclusion

This project successfully developed a comprehensive Mutual Fund Analytics Platform that integrates data engineering, financial analytics, business intelligence, and dashboard visualization into a unified solution.

The project began with the ingestion and validation of multiple mutual fund datasets, followed by data cleaning, database design, exploratory analysis, performance evaluation, advanced analytics, and dashboard development. Key financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR were implemented to assess fund performance and risk characteristics.

The resulting Power BI dashboard provides an interactive environment for exploring industry trends, fund performance, investor behavior, and market dynamics. Additionally, advanced

analytical techniques such as cohort analysis, SIP continuity analysis, and portfolio concentration assessment generated actionable business insights.

Overall, the project demonstrates the practical application of data analytics techniques within the fintech domain and highlights how structured data-driven approaches can support investment decision-making, investor engagement, and business intelligence initiatives within the mutual fund industry.